MNIST Handwritten Digit Classification POC

Overview:
This project explores how handwritten digits can be recognized using the well-known MNIST dataset. The goal was to test whether careful data analysis and model selection could lead to strong results, while also keeping the approach efficient and easy to explain.

Dataset and Preparations:
The dataset contains 70,000 grayscale images of digits (0–9). Each image is 28×28 pixels, which gives 784 values per digit. Before training, the images were normalized (0–255 → 0–1), and exploratory data analysis (EDA) was used to understand which parts of the images carried the most useful information.

Eda insights:

- Most of the important pixels are in the center of the image, while the borders carry little information.
- The dataset is balanced, so no extra class adjustments were needed.
- Some digits (like 4 and 9, or 3 and 8) are naturally harder to separate and often confused.

Models And Results:
Two different approaches were tested:

1. Random Forest (Traditional ML)
    - Trained on high-variance pixels only
    - Achieved 94.1% accuracy
    - Fast and interpretable, but less powerful
2. Convolutional Neural Network (Deep Learning)
    - Used a simple 3-layer CNN with dropout
    - Achieved 98.9% accuracy
    - Learned image features automatically and performed much better

Model	Accuracy	Training Time

Random Forest	94.1%	~2 minutes
CNN	98.9%	~15 minutes

Learning Outcomes:

- The CNN clearly outperformed the Random Forest, showing the strength of deep learning for image data.
- Feature selection helped reduce computation for the Random Forest without hurting accuracy.
- The confusion matrix confirmed the predictions from EDA about which digits would be hardest to classify.

If expanded further, the project could benefit from:

- Training for more epochs and adding data augmentation (rotation, scaling, noise).
- Fine-tuning CNN hyperparameters for even better results.
- Deploying the model through a simple API or web app.
- Adding explainability methods (like LIME/SHAP) to understand predictions better.

Conclusion:

This proof of concept shows that combining EDA insights with machine learning leads to strong, efficient models. The CNN achieved almost 99% accuracy, proving that this approach is highly suitable for real-world digit recognition tasks.